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S-, *P*-stabilized bromoallenes for modular allenylborylation of alkenes

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Simultaneously forming a carbon–carbon and a carbon–heteroatom bond in a single step through transition metal-catalyzed alkene difunctionalization strategy has emerged as a powerful tool for synthetic organic chemistry. Due to the uncontrollable reactivity, direct cross-coupling with bromoallenes as the building blocks for the selective allenation and borylation remains challenging. We herein report a new type of *S*- and *P*-stabilized bromoallenes for palladium-catalyzed modular allenation and borylation of alkenes to the divergent synthesis of multiply functionalized allenes in a highly regio- and diastereoselective manifold. The reaction features broad substrate scope and wide functional group compatibility, thus providing a straightforward method to install allenyl and boryl groups across alkenes. Control experiments highlight the crucial importance of *S*-, *P*-stabilization for the oxidative insertion of Pd-species into the allenyl–Br bond. The facile syntheses of bioactive allenic steroids and exocyclic allenes demonstrate the synthetic utility of this protocol.

alkene difunctionalization, olefins, bromoallenes, allenylborylation, palladium

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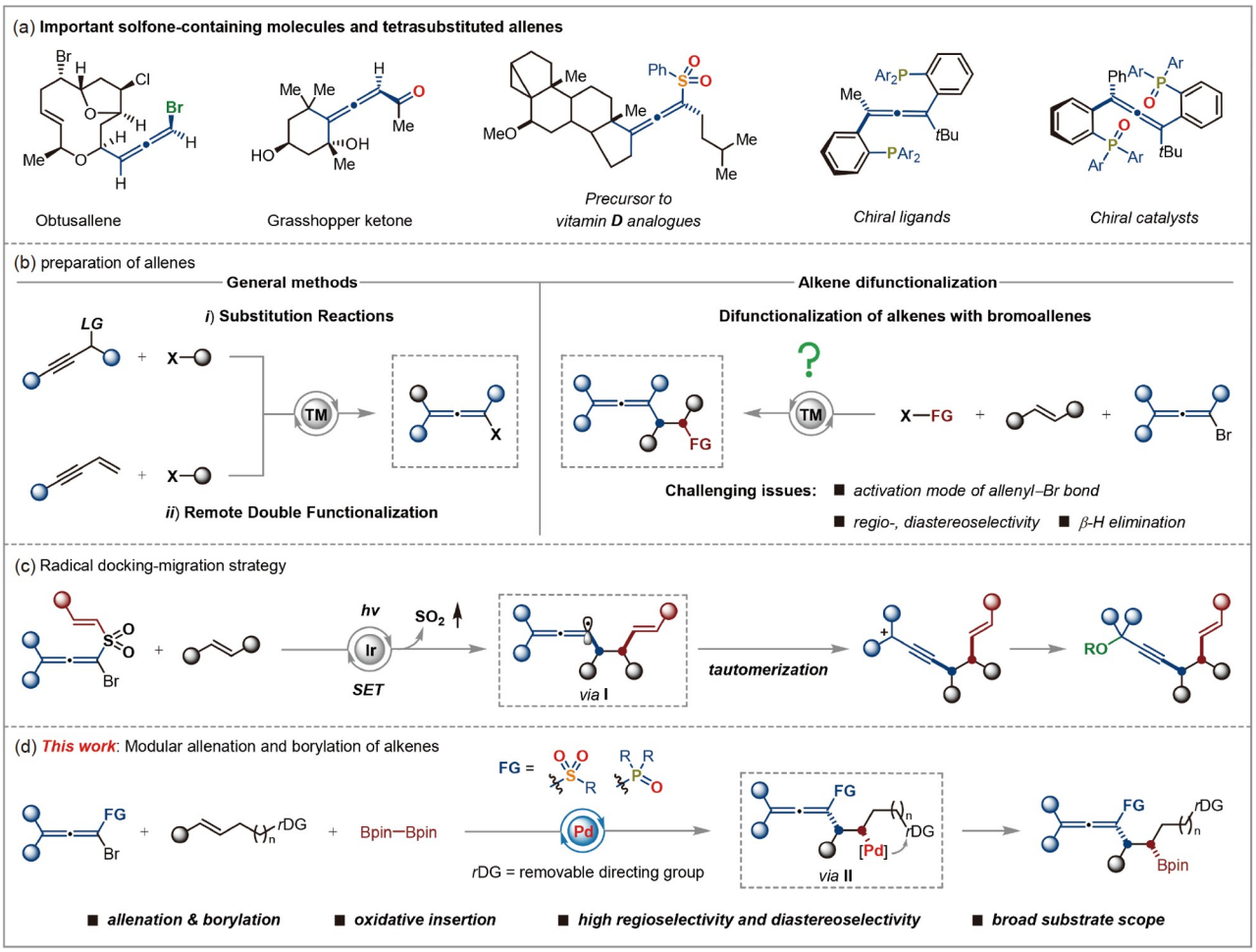
1 Introduction

Polyfunctionalized allenes are very important building blocks in organic transformations [1–3] and material science [4,5] due to their unique orthogonal cumulative π-systems. Many efforts have been made toward their syntheses by developing efficient and practical methodologies during the last decades [6–8]. Among them, multi-functionalized and exocyclic allenes bearing valuable functional groups, such as halides, sulfones, and phosphines, are highly desirable units and widely present in natural products [9,10], bioactive molecules [11,12], chiral phosphine ligands [13,14], and catalysts (Scheme 1a) [15]. Hence, there is a continued strong demand for the efficient and selective synthesis of such scaffolds in a diverse fashion.

Thus far, catalytic substitution reactions with propargylic derivatives [16–20] and remote double functionalization of 1,3-enynes have recognized as the synthetic tools for affording allenes with one or two synthetically useful functional groups [21–25]. On the other hand, transition-metal-catalyzed 1,2-difunctionalization of unactivated alkenes with two different functional groups has been recognized as an increasingly versatile tool for the synthesis of highly complex molecules [26–30]. The bench-stable bromoallenes

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Scheme 1 Background of the synthesis of functional allenes and current transformations of bromoallenes (color online).

[31], which are attractive alternative for direct allenation *via* transition-metal-catalyzed cross-coupling to the synthesis of decorated allenes in a straightforward manifold [32–36]. As to the transformations in alkene difunctionalization, we anticipated that once the alkene underwent allenylmetalation with a bromoallene, the *in situ* formed alkylmetal species would be coupled with a suitable nucleophile. However, several major challenges still need to be overcome: (1) while avoiding both C=C insertion and radical tautomerization of bromoallenes, the transition metal must selectively undergo oxidative addition into the allenyl–Br bond; (2) controlling the regioselectivity and diastereoselectivity; (3) β-H elimination of the Csp^3–[M] intermediate needs to be prohibited (Scheme 1b).

Very recently, Zhu *et al*. [37,38] disclosed a new type of allenyl bromide bearing alkenyl sulfone moieties for the radical-mediated 1,2-alkynylalkenylation and 1,2-enynylalkenylation of alkenes in the presence of photocatalyst under irradiation. Here, the sulfone group is critically important not only as a linker for intramolecular radical migration, but also for stabilizing the initial allenyl radical. Notably, the reformed allenyl radical **I** undergoes a facile tautomerization process, thus achieving sequential installation of alkynyl and alkenyl groups across alkenes (Scheme 1c). Beyond that, the process consisting of highly selective C=C insertion of bromoallene, along with a subsequent β-elimination affording a propargylic compound was also observed in manganese-catalyzed direct C–H propargylation of indoles [39]. These results suggested that the bromoallenes were preferential transferred into propargyl and alkynyl functionalities. As such, a practical approach for the construction of a new functionalized allenes *via* precise allenyl–Br bond activation is highly desirable.

With this purpose in mind, we became interested in developing an efficient alkene difucntionalization to the divergent synthesis of decorated allenes with bromoallenes. In order to stabilize the *in situ* formed allenyl-Pd intermediate, we herein designed and prepared a range of practical and bench-stable *S*- and *P*-stabilized bromoallenes, specifically for modular palladium-catalyzed allenylborylation of unactivated alkenes. Indeed, various bromoallenes bearing sulfones, phosphines and trifluoromethyl moieties react

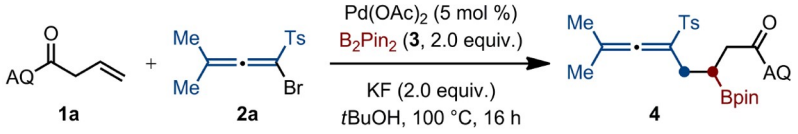
smoothly as the precursors of allenyl modules. The Pd-catalyst exactly undergoes oxidative insertion to the allenyl–Br bond and subsequent 1,2-migration to form the key intermediate **II** as a single isomer, which proceeds *via* sequential transmetalation and borylation across the double bond of olefins, thus achieving the first example of modular installation of allenyl functionality into unactivated alkenes with bromoallenes. It is worth noting that our protocol features a complete control of chemoselectivity, regioselectivity, and diastereoselectivity, as well as excellent functional group compatibility and broad substrate scope (Scheme 1d).

2 Results and discussion

We initiated our studies by optimizing the reaction conditions for the efficient allenyl installation into unactivated alkene **1a** with bromoallene **2a** as the electrophile and B_2pin_2 as the nucleophile through chelation-assistance of 8-aminoquinoline (AQ) [40–46]. To our delight, after surveying reaction conditions we found that 5 mol% of $Pd(OAc)_2$ furnished the desired allenylborylated product **4** in 98% yield when employing *t*BuOH as the solvent at 100 °C, the other representative mediums, such as tetrahydrofuran (THF), toluene, and acetonitrile (MeCN) gave significantly reduced yields (Table 1, entries 1–4). Moreover, the nature of various potassium salts, as well as trimethylamine (Et_3N) appeared to be crucial. Experiments identified KF as the ideal base of choice (entries 5–8). Interestingly, sharp contrasting results were obtained when using different Pd(0)-complexes as the metal source, while no reaction was observed in the absence of Pd-catalyst (entries 9–11). Moreover, lowering the reaction temperature to 70 °C resulted in a significantly reduced yield of **4** (entry 12). Thus, our optimization studies suggested that a simple catalytic system consisting of $Pd(OAc)_2$, a base of KF, efficiently enabled the regioselective allenylborylation of unactivated alkene (**1a**) with bromoallene (**2a**).

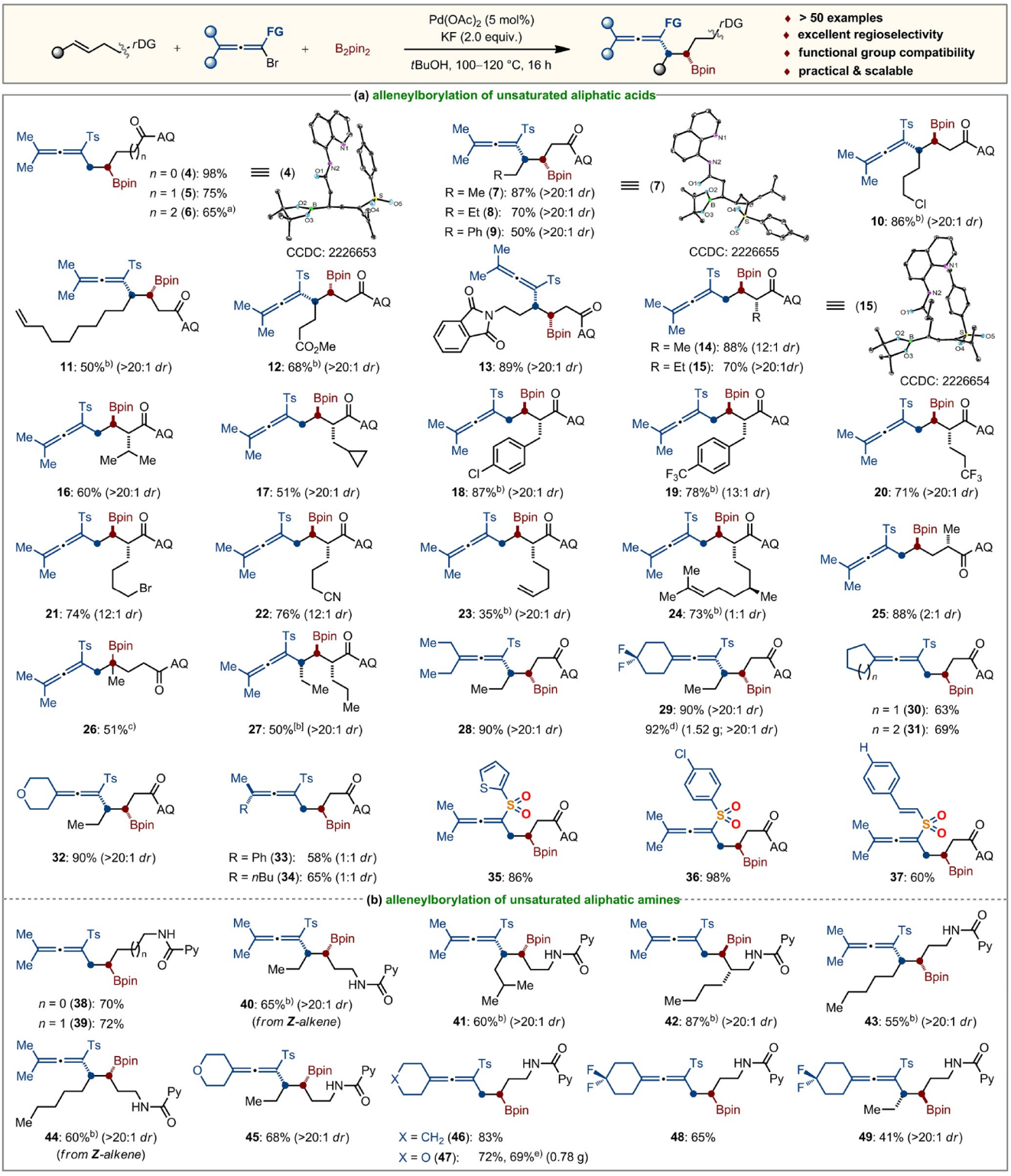
By the use of the sulfonyl-stabilized bromoallenes (**2**) as the allenyl precursors, we examined the generality of palladium-catalyzed alkene allenylborylation with B_2pin_2 (**3**) as the boron-based nucleophile (Scheme 2a). Firstly, 3-butenoic, 4-pentenoic as well as 5-hexenoic acids were employed as the suitable substrates for the envisioned allenylborylation to afford the corresponding products **4**–**6** *via* selective β,γ-, γ, δ-, and δ,ε-difunctionalization. Various internal alkenes bearing methyl, ethyl, and phenyl groups smoothly delivered the desired allenes **7**–**9** with moderate to good yields and excellent diastereoselectivity. Additionally, unsymmetrical olefins containing the more hindered aliphatic substituents bearing chloro, double bond, ester and phthaloyl-protected amine functional groups were well-tolerated under the reaction conditions, giving the polyfunctionalized allenes **10**–**13** in 50%–89% yields with complete control of diastereoselectivity. Remarkably, the allenylborylation also smoothly occurred for substrates containing sterically hindered α-substituents (**14**–**24**). A variety of valuable electrophilic functional groups, such as cyclopropyl (**17**), trifluoromethyl (**19** and **20**), chloro (**18**), bromo (**21**), nitrile (**22**), and alkenes (**23**, **24**) were well-tolerated by the palladium catalytic system, delivering the polyfunctionalized allenes in 35%–88% yields with high diastereoselectivity. For γ,δ-unsaturated carbonyl compounds bearing α- or γ-methyl group were well competent, leading to secondary or tertiary allenylated alkylboronates **25** and **26** in good yields *via* six-membered pallada-cycles. Notably, a more challenging internal olefin underwent allenation and borylation successfully to generate the product **27** which bears three chiral carbon centers with excellent control of diastereoselectivity.

Table 1 Optimization for allenylborylation of alkenes with **2a** [a]



Entry	Modified conditions	Yield (%) [b]	Entry	Modified conditions	Yield (%) [b]
1	none	98 (75) [c]	7	KOAc instead of KF	trace
2	THF instead of *t*BuOH	55	8	Et_3N instead of KF	70
3	Toluene instead of *t*BuOH	79	9	$Pd(PPh_3)_4$ instead of $Pd(OAc)_2$	20 [d]
4	MeCN instead of *t*BuOH	79	10	Pd_2dba_3 instead of $Pd(OAc)_2$	98 [d]
5	K_2HPO_4 instead of KF	77	11	Without Pd-source	0
6	K_2CO_3 instead of KF	21	12	Under 70 °C	40

a) Reaction conditions: **1a** (0.1 mmol, 1.0 equiv.), **2a** (1.5 equiv.), B_2pin_2 (2.0 equiv.), [Pd] (5 mol%), base (2.0 equiv.), solvent (2.0 mL), @100 °C, 16 h, under N_2. b) Isolated yields. c) Pd(OAc) (2.5 mmol%). d) [Pd] (10 mmol%).



Scheme 2 Substrate scope for the palladium-catalyzed allenation and borylation of alkenes with *S*-stabilized bromoallenes. Reaction conditions: alkene (0.1 mmol, 1.0 equiv.), bromoallene (0.15 mmol, 1.5 equiv.), B_2pin_2 (0.2 mmol, 2.0 equiv.), $Pd(OAc)_2$ (5 mol%), KF (2.0 equiv.), *t*BuOH (2.0 mL), @100 °C, 16 h. a) Detected by ^1H NMR analysis. b) @120 °C. c) $Pd(OAc)_2$ (10 mol%) was used, @110 °C. d) 2.5 mmol scale. e) 2.0 mmol scale (color online).

Thereafter, a series of acyclic, and exocyclic bromoallenes were also investigated for allenylborylation under the catalytic system. The desired products **28**–**34** were afforded in excellent yields, albeit with poor diastereoselectivity of compounds **33** and **34** when employing racemic bromoallenes as the electrophiles. As expected, conjugated

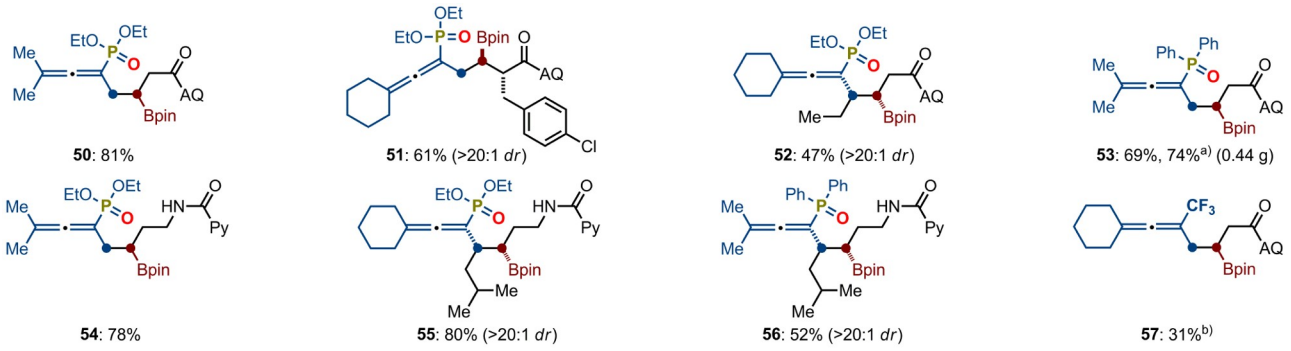
bromoallenes with different (hetero)aryl and alkenyl sulfones were proven to be viable substrates for the cascade allenylborylation reaction with alkenes to afford the corresponding products **35**–**37** in high yields.

Inspired by the success of our palladium-catalyzed subsequential allenylborylation of unsaturated aliphatic acid derivatives, we became intrigued by examining the possibility for direct allenation of alkenyl amines with the assistance of picolinamide auxiliary (Scheme 2b) [47–49]. Terminal homo- and bishomoallylamines were initially tested and delivered the corresponding allenylated amines **38** and **39** *via* five- or six-membered palladacycles. Both internal alkenylamines and terminal alkenylamines bearing an alkyl-substituent at the α-position proceeded in excellent regio- and diastereoselective fashion, affording **40**–**44** in 55%–87% yields. Likewise, the allenation of alkenylamines also exhibited high generality toward various exocyclic bromoallenes, thereby providing an expedient route to access aminoboryl-functionalized exoallenes **45**–**49** in moderate to good yields, as well as high diastereoselectivity.

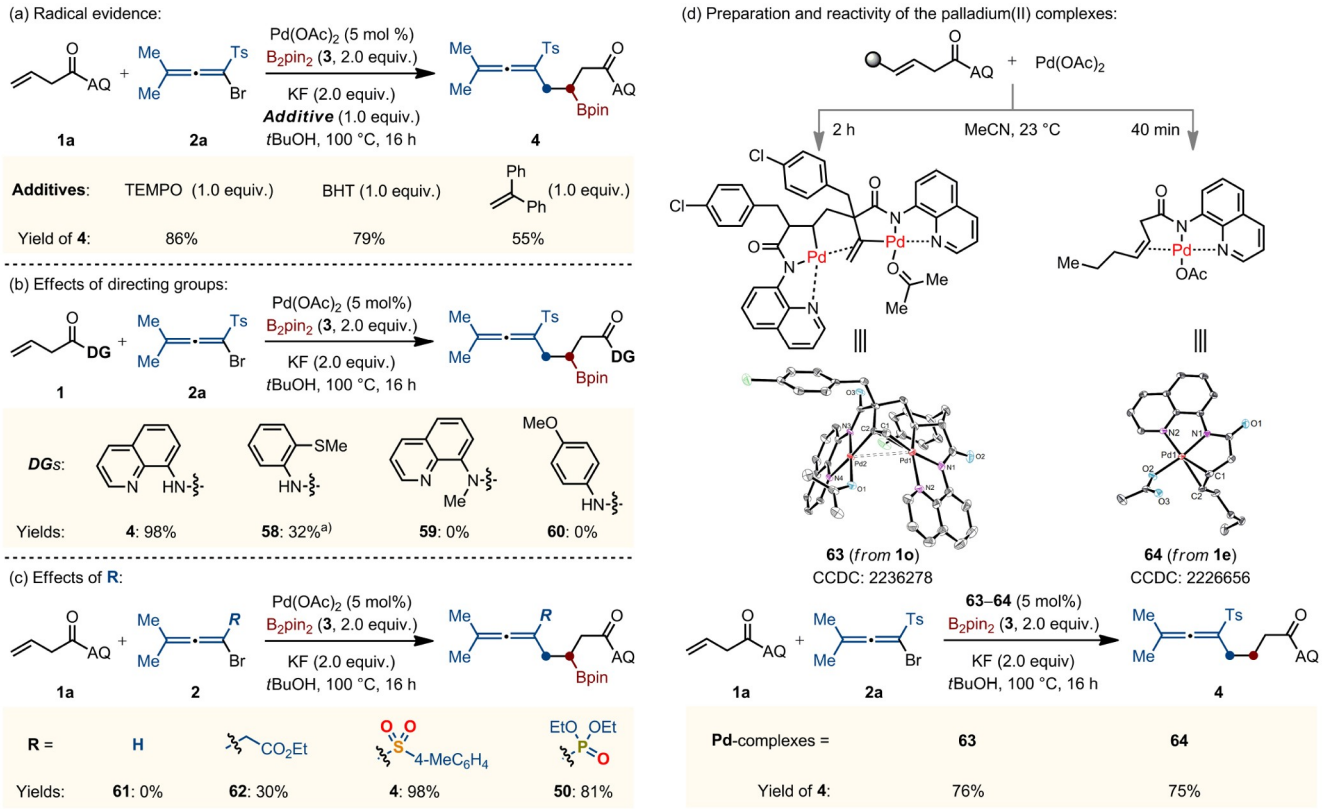
On the basis of the remarkably high efficacy of the alkene allenation with *S*-stabilized bromoallenes, we further investigated analogous allenylborylation of unactivated alkenes with different *P*-stabilized bromoallenes (Scheme 3). It is noteworthy that both phosphonate and phosphine oxide substituted bromoallenes were well-tolerated by palladium catalyst, successfully undergoing cascade allenylborylation and forming the synthetically useful allenyl phosphorous compounds **50**–**56** in a modular fashion. Moreover, the electron-deficient trifluoromethyl-substituted bromoallene was found to be a viable substrate as well, albeit with a rather modest yield of product **57**.

Regarding the reaction mechanism, a series of control experiments with stoichiometric quantities of representative radical scavengers, such as 2,2,6,6-tetramethyl-1-piperinedinyloxy (TEMPO), butylated hydroxytoluene (BHT), and 1,1-diphenylethene, were performed. The allenylborylation product **4** was obtained with negligibly reduced yields under the described conditions (Scheme 4a). These results strongly indicate that the reaction undergoes a non-radical process. Importantly, we also examined the envisioned alkene allenylborylation with differently substituted amides. The presence of a bidentate directing group, such as 8-aminoquinoline or 2-methylthioaniline, was proven to be crucially important for activating the palladium catalyst and promoting the regioselective allenylation and borylation (Scheme 4b). Intrigued by the unique effects of α-functionalities of bromoallenes, we prepared the bromoallenes bearing different α-substituents and performed the parallel experiments under the identical reaction conditions. No reaction or significantly reduced yield of **62** were observed when employing bromoallenes bearing α-H or CH_2CO_2Et substituents as the electrophiles. In sharp contrast, *S*- and *P*-stabilized bromoallenes delivered the corresponding products (**4**, **50**) in excellent yields (Scheme 4c). These observations highlighted the unique conjugated *S*- and *P*-stabilization for the oxidative insertion step of Pd-species into allenyl–Br bond. Moreover, to further unravel the mechanistic intricacies of the bidentate directing group, we prepared and isolated a Pd-dimer species of $[AQ]_2[Pd^{II}]_2$ (**63**), the formation of which might involve a intermolecular alkenyl C–H activation to a terminal double bond [50]. Beyond that, a $[AQ]Pd^{II}$ monomer **64** was also prepared under similar reaction conditions. Both of the structures were confirmed by X-ray crystal analysis. Notably, catalytic amount of these Pd^{II}-complexes were found to be catalytically active catalysts, as **4** was invariably formed in good yields (Scheme 4d). These observations suggested the crucial directing and ligand-tuning effects of the bidentate groups.

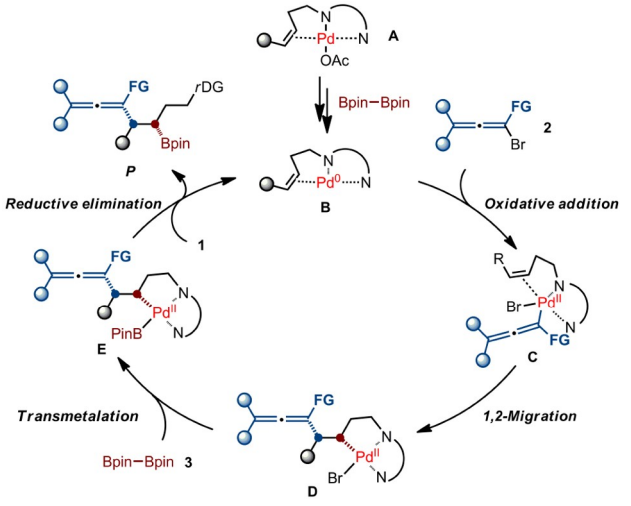
Based on the above findings and previous insights [43], a plausible catalytic cycle for palladium-catalyzed sequential allenation and borylation of unactivated alkenes was proposed in Scheme 5. The *in situ* formed Pd^{II}-species (**A**) was initially reduced by B_2Pin_2 to afford the catalytically active Pd^0-species (**B**), which selectively underwent an oxidative insertion to the allenyl–Br bond. A key allenyl–Pd^{II} intermediate (**C**) was generated with the assistance of sulfonyl or phosphoryl substituents. This intermediate **C** could undergo



Scheme 3 Allenylborylation with phosphinated and trifluoromethylated bromoallenes. Reaction conditions: alkene (0.1 mmol, 1.0 equiv.), bromoallene (0.15 mmol, 1.5 equiv.), B_2pin_2 (0.2 mmol, 2.0 equiv.), $Pd(OAc)_2$ (5 mol%), KF (2.0 equiv.), *t*BuOH (2.0 mL), @100 °C, 16 h. a) 1.0 mmol scale. b) @90 °C (color online).



Scheme 4 Mechanistic studies (color online).

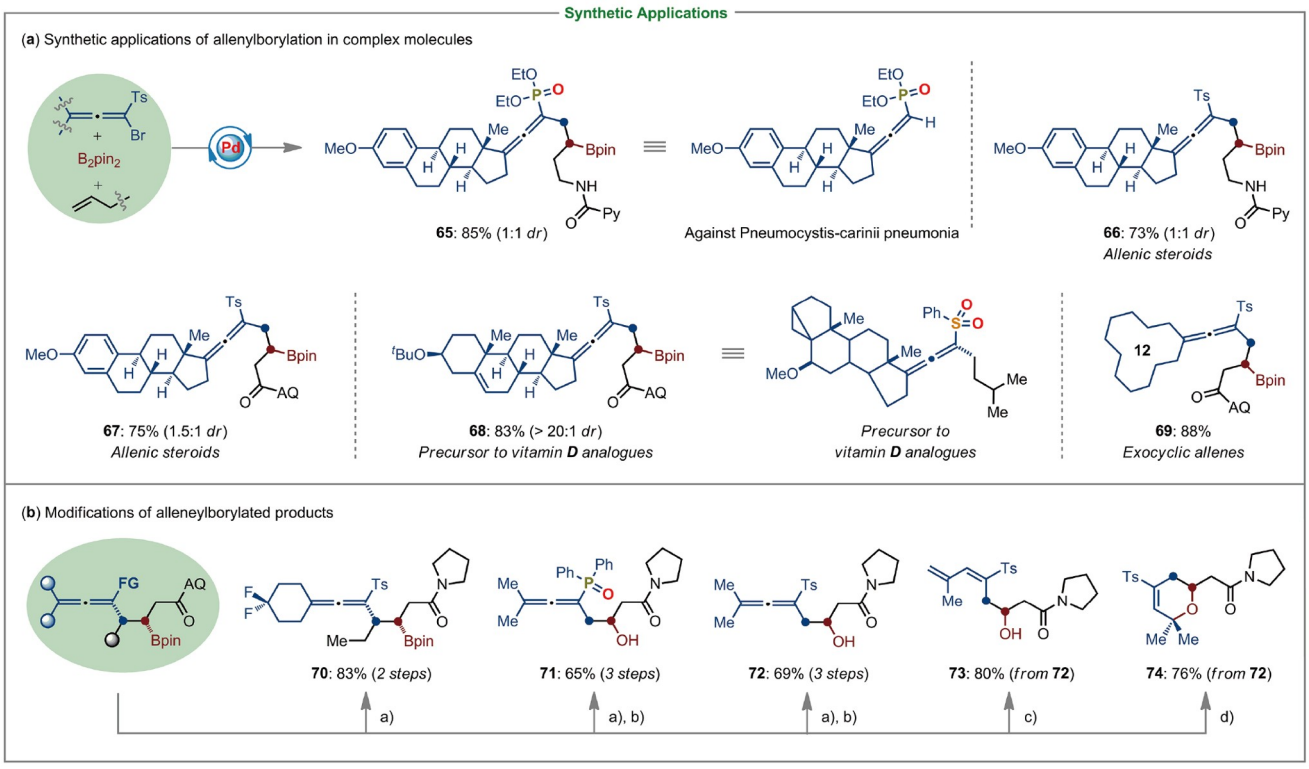


Scheme 5 Plausible catalytic cycle (color online).

an intramolecular 1,2-migration to form **D**, followed by a transmetalation reaction with B_2Pin_2 to generate intermediate **E**. Finally, a sequence of reductive elimination of **E** and re-coordination of Pd-species with another alkene (**1**) liberated the allenylborylated products and regenerated the active Pd^0-catalyst **B** (Scheme 5).

To demonstrate the versatility of palladium-catalyzed modular allenylborylation, we further synthesized the allenic steroids [11] **65**–**68** in 73%–88% yields. Among them, product **65** is an analogue to the reported steroidal allenic phosphonic acid derivative which displayed greater effects against pneumocystis-carinii pneumonia. Moreover, a precursor to vitamin **D** analogue [12] **68** could also be easily prepared under our reaction conditions. Beyond that, synthetically useful macrocyclic allenes, such as twelve-membered exocyclic allene **69**, can be easily prepared with excellent yield by our simple catalytic system. These results should prove instrumental for synthesis of structually complex molecules that enable chemical biology research (Scheme 6a).

Moreover, the synthetic applications of functionalized allenylborylation products were further illustrated by various facile transformations (Scheme 6b). Firstly, a reaction sequence consisting of Boc-protection and nucleophilic substitution removed the directing group of 8-aminoquinoline to aliphatic amides in high yields (**70**; 83% yield). Thereafter, the Csp^3–B bond can be easily transformed to Csp^3–O bond by oxidation reaction, giving the corresponding allenyl alcohols **71** and **72**. Finally, the allenyl scaffold of **72** could be further converted into highly *E*-selective 1,3-diene **73** in the presence of TfOH, while 15% of a Au-catalyst enabled intramolecular cyclization to afford pyran **74**.



Scheme 6 Synthetic applications of allenylborylation and modifications of alleneylborylated products. Reaction conditions: a) i) Boc_2O (2.0 equiv.), DMAP (10 mol%), MeCN, @60 °C, overnight; ii) pyrrolidine (1.5 equiv.), PhMe, @60 °C, overnight. b) $NaBO_3·4H_2O$ (5.0 equiv.), **6** (0.2 mmol), THF/H_2O, @23 °C, 12 h. c) i) TfOH (1.5 equiv.), DCM, @23 °C, 2 h; ii) 23 to −35 °C; iii) frozen HCl_{aq}, @−60 °C. d) $Au(PPh_3)Cl$ (15 mol%), $AgBF_4$ (15 mol%), THF, @23 °C, 3 h (color online).

3 Conclusions

In conclusion, we have reported the preparation of *S*- and *P*-stabilized bromoallenes, which proceeded oxidative insertion of Pd-species into allenyl–Br bond to form the key allenyl–Pd intermediates, thereby achieving modular allenation and borylation of unactivated alkenes in a regio-, site-, and diastereoselective fashion. Control experiments highlighted the crucial importance of the *S*- and *P*-conjugation for enhancing the reactivity of bromoallenes, as well as the unique directing and ligand-tuning effects of bidentate groups for ensuring chemo- and diastereoselectivity. Moreover, the synthetic utility of our alkene allenylborylation was well-illustrated by the facile syntheses of bioactive allenic steroids and derivatizations of the resulting allenes. We believe this work will open a perspective for the modular preparation of multi-functionalized allenes and find wide applications in allene chemistry.

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Conflict of interest The authors declare no conflict of interest.

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